

GBE in the UK Green Transformation Path Selection

Analysis Report

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1. Introduction

The world is facing a "triple planetary crisis" of climate change, pollution and biodiversity loss. Reports from the United Nations and the World Meteorological Organization show that climate change has already led to record levels of greenhouse gas emissions, more extreme weather events, melting glaciers and rising sea levels. These phenomena are having a profound impact on the Earth's ecosystems, with rising sea levels and an increase in extreme weather posing an unprecedented threat to global ecology and human society. When the tipping point of climate change is reached, it will bring irreversible disasters, especially the impact of increased extreme weather and sea level rise on coastal areas, which will further exacerbate socio-economic inequalities and affect living conditions in poor and ecologically fragile regions of the world. Therefore, there is an urgent need for countries around the world to take countermeasures, and the clean energy transition is considered a key strategy to solve the problem of climate change.

As one of the first countries in the world to commit to net zero emissions, the UK has actively responded to the challenge of global climate change and has set a target of achieving net zero emissions by 2050. In response, the UK government established the Great British Energy (GBE), a publicly owned energy company that will strive to develop clean energy. Not only greenifying the energy production, the GBE also aims at developing energy security, maintaining the bills low for the consumers, and modernizing and sustainable the UK's energy industry. However, achieving this goal will not be easy, and GBE faces many challenges along the way, especially in terms of how to maintain economic viability and social equity in the rapidly growing renewable energy sector.

The core objective of the GBE is to reduce carbon emissions by promoting the production and use of renewable energy. However, the success of the GBE depends not only on the rapid deployment of clean energy, but also on ensuring economic stability. Renewable energy projects often require significant up-front investment, and how to make large-scale investments while ensuring that energy prices are reasonable

and affordable is a major challenge that GBE must face. In this context, the GBE not only needs to achieve the transition to green energy, but also needs to strike a balance between energy equity, stability and reasonable prices to ensure that consumers, especially low-income households, can benefit from it.

The purpose of this report is to provide policy advice to the UK Minister for Energy Security and Net Zero Carbon Emissions on the future development of GBE by analysing the financial, social and environmental performance of the three state-owned energy companies EDF, Vattenfall and Ørsted. Through in-depth analysis of examples from domestic and international energy companies, this report aims to provide practical policy recommendations to help the GBE overcome its current challenges in driving the green energy transition and provide a sustainable path forward for the UK's energy future.

2. The Historical Evolution of the UK Energy System

2.1 Background to the Public-private Transformation in the 1980s

The oil shocks of the 1970s exposed the vulnerability of the UK's energy supply, in particular the high dependence of the UK energy system on imported energy. The surge in oil prices, especially the oil crises of 1973 and 1979, led to significant increases in global energy prices. As it is heavily reliant on imported oil and gas, the UK's energy prices and supply are greatly affected by the instability in the energy markets worldwide. This led the UK government to acknowledge that the energy policy needed to drastically change. With the arrival of the Thatcher government in 1979, the government initiated a series of market-oriented reforms that would make energy supplies more efficient and reliable by reducing the government's role in the energy sector and easing the fiscal pressures.

During the government of Prime Minister Margaret Thatcher, the United Kingdom pursued a large-scale reform of privatization, prioritizing the privatization of the

energy sectors like electricity, gas and coal. During the early 1980s, the government encouraged competitiveness and efficiency in the marketplace by selling several government-owned energy firms (e.g., British Electricity, Gas, etc.) to private enterprises. According to the government, the introduction of privatization will bring about competition, enhance the efficiency level in the energy sector, and enable the flow of capital, easing the load on the government's finances. Through these changes, the government expects the economy to grow while lessening fiscal pressures, and expects that this will provide the opportunity for stimulating competitiveness and innovations within the energy sector.

However, the privatization reform has brought many negative effects. First, after privatization, the profitability targets of energy companies became dominant, leading to higher energy prices, especially for low-income households, which meant significantly higher energy bills. The rise in energy prices after privatization of about 30%, especially during the cold winter months when energy demand increases, is highly volatile, further exacerbating social inequalities. In addition, the privatization reform also raises the question of the security of energy supply. As a result of market reforms, the stability and long-term reliability of energy supply have been affected. The volatility of energy prices and the high dependence on imported energy have made energy security a prominent political and economic issue. As a result, Britain's energy system is losing control and stability. The long-term dependence on market forces for energy supply makes the government powerless to deal with the energy crisis (Pearson & Watson, 2012).

2.2 Further Liberalization and Transformation of Energy Markets

(1990s - 2000s)

In the 1990s, the British government continued to liberalize the energy market. Through the reform, the government hopes to further improve the efficiency of energy supply and promote the development of clean energy by introducing more market

competition. However, the liberalization of the energy market is also accompanied by upward pressure on energy prices, especially during the winter peak demand period. The government has gradually liberalized the energy market, allowing more private companies to enter the market in order to make the market more competitive and improve efficiency. At this time, the awakening of environmental awareness prompted the government to pay attention to the low-carbon transition, and gradually make green energy as one of the core of the future energy strategy. Green energy policy gradually gained attention in the 1990s, and the government began to propose how to achieve the low-carbon transition of energy and set the goal of gradually developing renewable energy. Clean energy, such as wind and solar energy, has gradually entered the government agenda and received policy support. By 2000, the government had enacted the Climate Change Act, setting emission reduction targets for 2020 and 2050, which laid the foundation for green energy policies (Steinfort, Rowan, 2024).

However, although green energy policies have been promoted by the government, technological limitations and market limitations have made renewable energy relatively expensive in the initial stage. Wind and solar cost much more than traditional energy sources, and because of their intermittency and instability, it is difficult to completely replace traditional energy sources. According to the IPCC (2023), although wind and solar energy gradually increased in the late 1990s, their share in the energy supply remains low. In this context, the government has chosen to encourage the development of green energy by providing subsidies and policy support, and to reduce the production cost of renewable energy through financial subsidies (IPCC, 2023).

At the same time, the further liberalization of energy markets has led to new challenges. The instability of energy prices and excessive competition in the market, especially during peak energy demand, have led to sharp fluctuations in energy prices. Although the government has promoted the development of low-carbon energy

through green energy policies, the market's acceptance of green energy and the maturity of technology still face great challenges. These problems have prompted the government to gradually adjust the policies of the energy market and begin to emphasize how to balance the role of market forces and public policies in the energy transition to ensure the stability and fairness of energy supply (Steinfort, Rowan, 2024).

2.3 Background and Historical Continuation of GBE

After several rounds of market-oriented reforms and green Energy transition, the British government established the Great British Energy (GBE), marking a major transformation of the British energy system. The core objective of the GBE is to drive the transition to green energy through public control and, in the process, reduce energy bills for consumers. Unlike previous privatization reforms, the establishment of the GBE is based on a more active state intervention in the energy system, especially investments in the clean energy sector (Pearson & Watson, 2012).

The establishment of the GBE can be seen as a continuation and deepening of the history of British energy policy. Over the past few decades, the UK government has promoted the market-oriented and green transformation of the energy sector through a series of policies, but these reforms have not fully addressed the issues of energy price volatility, market instability and energy equity. The GBE was established to ensure the stability of energy supply through publicly owned means and to promote the green transition, especially the development of renewable energy sources (Gibbs, 2024).

However, the practice of GBE faces many challenges. How to ensure that the GBE is not overly dependent on government funds while carrying out the green transformation is a major challenge for the GBE. The funding source of GBE mainly depends on the support of the government, but how to balance the relationship between the investment of government funds and the economic self-financing of GBE still needs to be considered. Secondly, how to balance the environmental goals and economic benefits of GBE is also a key issue in its development. Despite being

responsible for steering the changes required for the green transition, the GBE also faces the challenge of ensuring that consumers' bills do not rise while maintaining economic benefits (Steinfort & Rowan, 2024).

3. Overview of the Current UK Energy System

3.1 Analysis of Main Energy Sources and Suppliers

3.1.1 Energy Structure

Currently, the three forms of energy mainly used in the UK are gas, nuclear and renewable sources of energy. In the past ten years, the contribution of non-hydro electricity generation by renewable energy has gradually bisected, and the renewable energy sources which have shown an increase from year to year are wind and solar energy. However, despite the gradual increase in the share of renewables in the energy mix, it still faces problems of stability and price volatility. Gas is an important energy source within the United Kingdom, even though it has faced several challenges, such as the security of supply and rising prices. Nuclear also forms a significant portion of the total installed capacity with EDF, which is a procurer of nuclear supply for most of the nuclear power stations in the UK.

Gas is one of the UK's most widely used types of energy supplies today, contributing to approximately 40% of the country's total energy supply. However, the demand for natural gas has faced a problem of supply stability since it relies on imported energy resources, and the price of natural gas has tended to be unstable, raising the concern of energy supply risk. Thus, despite natural gas being a transitional fuel, its overreliance, in the long run, might have adverse environmental effects that are not so fitting, especially considering that the world is on a low-carbon transition path (Bolton & Hinson, 2020).

Nuclear power is still considered relevant in the energy mix of the UK, particularly when viewed in terms of its contribution to the fight against carbon emissions; nuclear

power is one of the low-carbon energy sources. Approximately 15 per cent of electricity is generated from nuclear power, which is very important if the stability of a country's energy supply and CO₂ emissions have to be curbed. Through these nuclear energy supply agreements, EDF determines the management of nuclear power in the UK, and due to the firm's nearly monopolistic power in the nuclear energy sector in the UK, it plays a very central role in the overall energy consumption (Pearson & Watson, 2012). In recent years, the UK government has strongly promoted investment in renewable energy, especially wind and solar. According to the CleanTechnica (2025), the installed capacity of wind and solar is increasing year on year, with the production of wind power increasing from 1.3 TWh in 2010 to 24.6 TWh in 2020. Solar energy increased from 0.5 TWh in 2010 to 4.7 TWh in 2020. The rapid growth of renewable energy has reduced the dependence on traditional energy sources to some extent, but still faces challenges from price fluctuations, technological developments, and energy storage issues (Steinfort & Rowan, 2024).

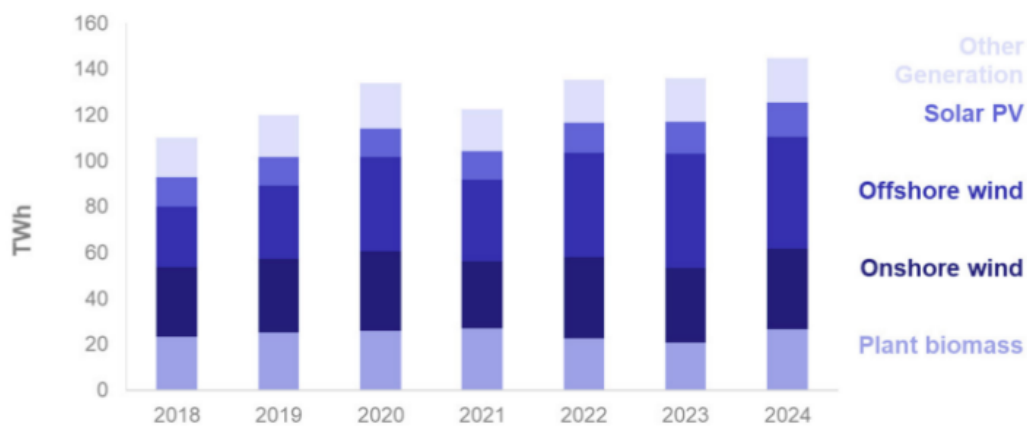


Figure 1: Renewable electricity generation in the UK by technology, 2018-2024

Source: CleanTechnica (2025)

3.1.2 Major Energy Suppliers

Currently, major Energy suppliers in the UK include companies such as EDF, Vattenfall, Ørsted, British Gas and Octopus Energy. Each company's role in energy

supply is different, and the following will look at the role and impact of each of these major suppliers.

➤ **EDF**

EDF is one of the UK's largest nuclear suppliers, operating most of the country's nuclear power stations, including major nuclear projects such as Hinkley Point C and Sizewell B. EDF provides around 15% of its electricity supply through its nuclear facilities, which is important in ensuring the stability of energy supply and the low-carbon transition. However, challenges facing EDF include high construction costs, project delays and market volatility. The Hinkley Point C project, for example, is already £1.8bn over budget and has faced construction delays. In addition, EDF is involved in green energy investments, such as offshore wind and solar projects (Pearson & Watson, 2012).

➤ **Vattenfall**

Vattenfall is a Swedish energy company that has actively invested in the renewable energy sector in recent years, especially in the wind energy sector. Vattenfall is committed to driving the global energy transition, especially in Europe, where Vattenfall's large-scale wind projects such as the Array wind farm in London, East of England, have made a significant contribution to the development of green energy in the UK. Vattenfall aims to generate more than 50% of its total energy production from renewable sources by 2030, demonstrating the company's proactive approach to the green transition (Steinfort & Rowan, 2024).

➤ **Ørsted**

Ørsted is a Danish energy company leading the offshore wind market worldwide. Ørsted also has extensive investments in the UK, particularly in offshore wind projects. Ørsted's Hornsea One wind farm is one of the largest offshore wind farms in the world and is important to the UK's green energy transition. Ørsted's strategy is focused on driving the global low-carbon transition through offshore wind projects and plays a vital role in achieving renewable energy targets (Pearson & Watson, 2012).

➤ **British Gas**

British Gas is one of the largest energy providers in the United Kingdom, primarily providing gas and electricity services. According to the report, British Gas occupies an important position in the UK's energy market. Although British Gas relies primarily on natural gas, it has also made efforts to drive the adoption of renewable energy, offering green power options to meet consumer demand for low-carbon energy. However, the challenge for British Gas is how to remain competitive in a gas market with volatile prices and gradually transition to renewable energy supply (Britishgas, 2024).

➤ **Octopus Energy**

Octopus Energy is a company that has emerged in the UK energy market in recent years, focusing mainly on the supply of renewable energy. Through its innovative business model, Octopus Energy leverages advanced smart grid technologies to deliver more flexible and sustainable energy solutions. Octopus Energy has established itself as one of the leading green energy suppliers in the UK, with investments in wind and solar projects in particular giving it a foothold in the market. Octopus aims to become the world's leading green energy company and drive the green transformation of the energy market through smart technology (Octopus energy, 2024).

3.2 Energy Market Challenges and GBE Opportunities

3.2.1 Diversity of Energy Supply and the Challenge of Price Volatility

With increased instability in global energy markets, the security of the UK's energy supply faces new challenges, particularly during the peak winter months. Global energy price volatility, international market uncertainty and geopolitical factors make the UK's energy security particularly vulnerable. Natural gas is one of the main energy sources in the UK, and its supply instability has posed a threat to energy security. In the winter peak demand, energy demand increases rapidly, how to ensure a stable energy supply has become an urgent problem for the government to solve.

Moreover, although the proportion of renewable energy is increasing year on year, the intermittency of wind and solar power remains a bottleneck in the UK power system. Renewable energy sources cannot provide consistent and stable power, especially during periods of insufficient wind or short sunshine, and power systems that rely on renewable energy are prone to instability. This implies that conventional energy, such as nuclear and natural gas, continues to guarantee and remain critical in ensuring energy supply. One of the issues that make markets in the UK particularly volatile is the cost and fluctuation of renewables (Pearson & Watson, 2012).

3.2.2 Opportunities Arising from Diversified Energy Supply and Energy Equity

Thus, being a publicly owned energy company, GBE has numerous advantages in advancing green energy while ensuring an adequate diversification of energy supplies with the help of state support. With regard to those strategies, GBE can use its status as a public enterprise that is yet to be privatized in its favour to diverse and secure energy sources as it implements renewable energy. In addition, with the integration of wind, solar, and conventional energy sources, GBE can ensure the proportion of diversified energy and decrease the reliance on imported energy, which, in turn, boosts the stability of the energy supply. GBE can invest more in energy storage to cope with the fluctuations of renewable energy simultaneously (Bolton & Hinson, 2020).

Besides flexibility measures, Gbes has an opportunity to enhance equity in energy supplies. Fuel poverty is predominantly observed in the UK because low-income families cannot pay the energy prices. Compared with state-owned enterprises, GBE could pay more attention to social equity when developing electricity price strategies. It will not frequently adjust the energy prices when the economic environment turns terrible. The capacity of the GBE to keep the prices affordable by shielding low-income groups from market fluctuations by capping laid down prices is important in handling fuel poverty in low-income households (UK Parliament, 2025).

However, one of the most significant problems of the GBE is the limited amount of public money to finance it. Despite this, the GBE relies on government subsidies. The over-dependence on this sort of funding might pressure the company, especially when there is high capital cost at the outset of the green transition. The question of how the GBE could advance to a green status without relying on excessive state aid and with a focus on its financial stability is one of the areas in its further development that must be solved (Adekoya et al., 2021).

4. Financial Performance Analysis of EDF, Vattenfall and Ørsted

4.1 Analysis of EDF's Financial Performance

EDF group is one of France's largest power utility companies, and many factors have significantly impacted its financial situation, such as Nuclear power project overruns, cost and external environmental forces. EDF has recorded fluctuating financial performance in the last few years mainly due to its enormous investment in Hinkley Point C and Flamanville nuclear projects, which affected its profitability. Net losses were recorded in 2019 and 2020, respectively. More specifically, EDF's losses in 2022 stood at 17.9 billion euros due to lower nuclear energy generation and significant write-offs (Le Monde, 2024). Thus, the financial profit of EDF in 2023 was 1 billion euros, which indicates its improved financial status. This profit is mainly due to the resumption of nuclear power production, especially in France, the significant increase in domestic nuclear power production. In 2023, EDF's nuclear power production was 320.4 TWh, at the top end of expectations, while in 2022 production fell to 279 TWh, the lowest level in three decades, due to corrosion problems and maintenance work at nuclear reactors (Reuters Staff, 2025).

Despite the recovery in profits, EDF's financial burden remains heavy. According to its 2024 financial results, EDF's sales were 139.7 billion euros, down 2.6% year-on-year. In this market environment, EDF's EBITDA (earnings before interest, tax,

depreciation and amortization) was €3.65 billion, showing that it has achieved some operating profitability from its nuclear and renewable energy projects. However, EDF still faces a high level of debt, with net financial debt of €54.3 billion at the end of 2024, and while it has stabilized compared to 2023, its debt-to-EBITDA ratio remains high, which poses challenges for the company's future financing ability (EDF, 2025). EDF could face several key risks to its future financial performance, starting with continued delays and cost overruns on the Hinkley Point C project. Although EDF plans to start construction in 2027, significant uncertainties remain regarding the project's construction schedule and cost due to funding issues, market uncertainties and construction delays. If the project is not completed on time or over budget, this will seriously affect EDF's long-term profitability and may even affect its overall financial stability (Monde, 2024). Second, EDF's high debt is a financial challenge. The company has accumulated a significant amount of debt over the past few years, and although it has been alleviated through asset sales and debt management, in the current high interest rate environment, future financing costs may increase further, which will increase the capital burden of EDF and affect the optimization and sustainability of its capital structure. Finally, market volatility continues to have a significant impact on EDF's financial results, in particular the instability of energy prices. While EDF's nuclear and renewable energy projects have recovered somewhat in 2023, price volatility in global energy markets can lead to sharp fluctuations in electricity selling prices, affecting the company's revenues and earnings. Especially during the winter demand peak, the rise in energy prices may further increase the pressure on EDF to compete in the market. Therefore, EDF needs to effectively deal with these risks to ensure that it can maintain sound financial growth in the future (Reuters Staff, 2025).

4.2 Vattenfall Financial Performance Analysis

As one of the largest energy suppliers in Sweden, Vattenfall's financial performance is relatively stable. In 2024, Vattenfall's total sales amounted to SEK 245.6 billion (€11.87 billion), and despite a decline from 2023, the company's profitability remains

strong. Vattenfall's EBITDA was €6.08 billion, reflecting its strengths in wind and hydropower (Buli, 2024). This stable revenue stream has enabled Vattenfall to maintain good financial health during the green transition.

Vattenfall has been actively expanding its renewable energy business in recent years, especially wind and hydropower. Over the past few years, Vattenfall has significantly increased its installed wind power capacity, which has provided it with a stable revenue stream. In 2024, Vattenfall's net profit was SEK 3.34 billion (about €320 million), although this figure is lower than in 2023, its profitability remains at a high level (Vattenfall, 2024).

However, one of Vattenfall's biggest challenges is the volatility of energy market prices. Especially during the winter peak, the sharp rise in energy demand often leads to a surge in electricity prices, which has a large impact on Vattenfall's profits.

Despite the increasing investment in renewable energy projects, the company's risk management in energy price fluctuations is still inadequate (BBC News, 2024).

Vattenfall's future financial performance will be influenced by a number of key factors. The first is the return on investment of renewable energy. While the wind and hydro businesses already provide Vattenfall with a steady stream of revenue, the high cost of construction and subsequent maintenance of these projects could affect profitability in the coming years. Wind power projects, in particular, have long-term revenue potential, but high construction costs and maintenance difficulties will be important challenges for companies (Vattenfall, 2024). Second, the volatility of global energy markets poses another risk to Vattenfall's financial performance that cannot be ignored. Volatility in energy prices, especially during peak winter demand, could put pressure on earnings. If energy prices fluctuate significantly, Vattenfall's revenues and profits will be directly affected, especially during periods of low prices, which may compress its profit margins (Buli, 2024). Finally, climate change and policy changes will also bring uncertainty to Vattenfall's long-term financial planning. While Vattenfall is driving a green transition and actively investing in renewable energy projects, the impact of climate change and changes in government policies could have a profound impact on the company's strategic implementation and financial targets.

For example, changes in energy policy could lead to changes in subsidies or new environmental regulations, which could affect Vattenfall's profit model and market performance. Therefore, Vattenfall needs to be flexible in responding to these changes to ensure its sustainable financial growth (Lawson, 2024).

4.3 Financial Performance Analysis of Ørsted

Ørsted is one of the world's leading renewable energy companies, particularly in the offshore wind sector. According to its 2024 financial report, Ørsted's total sales were 11.87 billion Dan klang (about 1.68 billion euros) and its EBITDA was 3.2 billion Dan Klang (about 430 million euros), showing its strong performance in the green energy sector. Ørsted's profitability is solid, especially driven by offshore wind projects, and the company's financial position continues to improve (Ørsted, 2024). The net profit of Ørsted was DKK 640 million (EUR 90 million) in 2024. Even though it was lower than in 2023, it was primarily due to project impairments and funding fluctuations. Projects such as Hornsea One is an offshore wind farm project being developed by Ørsted, and there are some of the largest offshore wind farms globally producing most of the company's total revenues.. However, as the company further expands its investment in offshore wind, the construction cost and technical challenges of the project will also become key issues for it (Millard, 2025).

Ørsted's financial performance will be influenced by a number of factors in the coming years. First, the return on investment of offshore wind is a key factor. Ørsted's strengths in the field of offshore wind are clear, and with its leading technology and strong market position, the company has made significant achievements in this field. However, the construction costs and technical challenges of offshore wind projects cannot be ignored. The high initial investment and complex maintenance requirements of these projects may affect Ørsted's profitability in the coming years, especially during the project construction phase and initial operation, which may lead to insufficient returns in the short term. Secondly, the volatility of global energy prices remains a significant risk for Ørsted. While Ørsted's main business is dependent on renewable energy, price instability in the global energy market,

especially in the context of high price volatility in the international electricity market, could still weigh on its profitability. For example, a sharp rise or fall in energy prices can affect their contract pricing and long-term returns, which in turn affects financial performance (Ambrose, 2024). Finally, policy changes and market access will also affect Ørsted's future financial performance. As the global green transition progresses, governments' support policies for renewable energy are likely to change. In particular, Ørsted may face policy uncertainty in terms of subsidy policy adjustments, changes in tax incentives, etc., which may affect its investment decisions and profit model (Jacobsen & Carlsson, 2025). Therefore, Ørsted needs to respond flexibly to changes in policy and market conditions to ensure its sustainable financial growth.

4.4 Comparative Financial Analysis of the Three Companies

For financial comparisons, we have chosen the following uniform financial measures: sales, EBITDA, net financial debt, net profit, capital structure. Here's how the three companies compare on these financial metrics:

Financial Metrics	EDF (France)	Vattenfall (Sweden)	Ørsted (Denmark)
Revenue	€139.7 billion (2023)	SEK 245.6 billion (€11.87 billion)	DKK 11.87 billion (€1.68 billion)
EBITDA	€3.65 billion (2024)	€6.08 billion	DKK 3.2 billion (€430 million)
Net Financial Debt	€54.3 billion (2024)	SEK 5.8 billion (~€540 million)	DKK 5.8 billion (~€780 million)
Net Profit	€1 billion (2023)	SEK 3.34 billion (~€320 million)	DKK 640 million (~€90 million)
Capital Structure (Net Debt/EBITDA)	1.49x	2.1x	1.8x

The data in the above table are from the latest financial reports of the three companies in 2024. As can be seen from the comparison table, the gap between EDF and Vattenfall is large in terms of sales and EBITDA. EDF's sales are relatively high, but its profitability is slightly lower due to high debt and project costs. Vattenfall's EBITDA is high, but its debt levels and market volatility remain risks for the future. Although Ørsted's sales are relatively low, its strategy of focusing on offshore wind

power has kept its market share and earnings performance in the green energy sector stable.

Overall, EDF has a strong investment performance in the nuclear and renewable energy sectors, but its financial performance is highly uncertain due to debt pressure and the delay of large-scale projects. Vattenfall's green transformation progress is relatively stable, but it is greatly affected by market fluctuations; Ørsted, for its part, has demonstrated a strong technological edge in the offshore wind sector, which is likely to continue to grow steadily in the future.

5. Social and Environmental Responsibility Analysis

5.1 EDF

EDF has faced a number of challenges regarding its social and environmental responsibilities, not least the controversy related to the Hinkley Point C project. Judging from recent news reports, EDF has faced allegations of unfair pay for its workers, and these issues have attracted widespread attention. According to the BBC, construction workers at Hinkley Point C project raised "unfair pay" claims, which triggered a standoff between the union and management, demanding that the company adopt a more equitable pay policy to avoid labor conflict (Bullock, 2024). In addition, EDF has faced widespread criticism as it pushes forward its plans for a "new nuclear age." According to Greenpeace, EDF's cost estimates for the nuclear project were overly optimistic, leading to significant underestimates of future funding needs and profitability. Hinkley Point C was supposed to be the new nuclear standard, but its spiraling construction costs, and delays have put the project at significant financial risk (Greenpeace, 2024).

In addition, EDF faces ecological and environmental challenges in its Hinkley Point C project. The Hinkley Point C Stakeholder Reference Group report also reveals the significant effects of the project on the environment of Wales, particularly the environment of the Severn Estuary. Among all the identified and analyzed risks of the

project, it is possible to regard the cooling water extraction system as the one that might have a critical impact on the fish population, including endangered species such as Twaité Shad. Highlighted in the report is the lack of adequate consideration for the cooling water extraction scheme put forward by EDF to have a fly-on into the environment should there not be sufficient protection of the environment (Hinkley Point C Stakeholder Reference Group, 2021).

From the environmental responsibility view, the issues challenging EDF are not only project-related factors such as delay and cost control. Instead, it also includes the impartiality and neutrality of the EDF in implementing environmental responsibility. To this end, the Greenpeace report revealed that EDF has not been transparent and has been audited several times concerning the nuclear power project, which might lead to a crisis of credibility regarding its claims about sustainability (Greenpeace, 2024). However, there is still ignorance of specific long-term ecological effects, such as how the C's cooling water system can affect the aquatic system; hence it is taken more comprehensive measures to mitigate which ensure that the company do not violate any environmental protection law and regulations plus international ecological agreements (Hinkley Point C Stakeholder Reference Group, 2021).

5.2 Vattenfall

As one of Sweden's largest energy suppliers, Vattenfall continues to demonstrate a high level of commitment to driving the green energy transition, particularly in wind and water. In recent years, however, the company has run into controversy over its social and environmental responsibilities, with some of its projects sparking widespread public protests and scrutiny from international environmental groups. First, Vattenfall's Clashindarroch Forest wind power project in Scotland sparked a massive social backlash. Wildcat Haven's social media campaign quickly gained traction in Sweden, with thousands of people protesting against Vattenfall on social platforms and pressuring the company to end the project by writing letters and other means. The construction of the wind farm is believed to threaten the extremely rare local wild cat population, particularly the endangered Scottish wildcat. Environmental

groups are concerned that THE habitat of these wild animals will be destroyed by construction, thus increasing the vulnerability of the ecosystem (THE, 2024). What is special about this campaign is that the pressure comes not only from the international community, but also from Vattenfall's own consumers, which has a clear impact on its brand image and customer base.

Second, Vattenfall is involved in the lignite mining business in Germany, which has long been criticized for its ecological damage. Although Vattenfall launched plans to sell its German lignite mines in 2014, claiming it was part of a transformation of its green strategy, the move has also raised questions from environmental groups.

Greenpeace Nordic even tried to buy the mines and shut them down permanently to meet its "keep coal in the ground" climate goal, although in the end Vattenfall chose other buyers. Environmental groups argue that Vattenfall has missed an opportunity to work with environmental groups to promote sustainable development by not really considering the opportunity for a just transition and prioritizing commercial interests instead (Fisher, 2022). The incident has exposed the social responsibility dilemmas Vattenfall faces in its green transition, particularly its ethical responsibility over how to deal with highly polluting assets.

In addition, Vattenfall's Norfolk Boreas offshore wind project in the UK has raised concerns about biodiversity conservation. Although the project has been hailed as an important part of the UK's goal of achieving net zero, the site is located close to the migration and habitat areas of endangered seabirds, which could have serious impacts on species including kittitoes and gannets. The Royal Society for the Protection of Birds (RSPB) pointed out that there is no mature mitigation plan for the deadly threat of wind power blades to birds, and the long-term ecological impact is difficult to assess. At the same time, 85 local parish councils have jointly opposed the project, arguing that the massive cable system required for construction will cause long-term disruption to residents' lives (Horton, 2021).

Despite Vattenfall's claim in the project plan that it "puts biodiversity at the heart" and its commitment to provide £15 million in compensation funds to local communities, these initiatives have failed to completely dispel public concerns about the

environmental impact. This reflects the need for Vattenfall to improve transparency and accountability in social communication, community engagement and ecological risk assessment as it pushes forward with renewable energy.

5.3 Ørsted

Ørsted's performance in social responsibility reflects its commitment to sustainability and social responsibility. However, on some issues, Ørsted still faces questions and challenges from the outside world.

First of all, Ørsted's achievements in offshore wind projects in recent years are notable, especially in the UK's Hornsea 3 project, where the company has driven the global renewable energy progress by building the world's largest offshore wind farm. However, in the process of achieving this goal, Ørsted faced industrial disputes with the unions. In October 2022, the RMT union, dissatisfied with Ørsted's pay package, organized strike action and, by sending a letter to the Danish Embassy, asked the Danish government to intervene and urge the company to renegotiate the pay issue with the union (RMT, 2022). Although the company made huge profits financially, the neglect of the pay and treatment of the employees led to confrontation with the union and had a negative impact on the company's image.

In addition, Ørsted faces some complex challenges in terms of environmental protection. According to its 2024 ESG report, Ørsted has taken several measures to reduce carbon emissions and enhance biodiversity conservation. The company is committed to achieving net zero emissions by 2040 and is promoting the construction of renewable energy projects through financial instruments such as green bonds and green loans in support of its global Energy transition goals (Ørsted, 2024).

Nonetheless, it should be noted that such initiatives have not been without controversy, especially when it comes to interacting with local communities, with some of Ørsted's projects, such as wind farms in the United States and Europe, encountering opposition from local communities and environmental groups. These

objections mainly focus on the potential impact of the project construction on local ecology and biodiversity, in particular the destruction of habitats for rare species.

5.4 Comparative Analysis of Environmental and Social Responsibility

When comparing the environmental and social responsibilities of EDF, Vattenfall and Ørsted, all three companies have made significant progress in driving the green energy transition, but they also face different challenges. EDF has faced worker compensation disputes and environmental risks in its nuclear energy projects, particularly in the Hinkley Point C project, where ecological impacts and industrial conflicts have become important issues (Bullock, 2024; Hinkley Point C Stakeholder Reference Group, 2021). Vattenfall's wind power project in Scotland has faced social protests, especially its impact on THE habitat of endangered species, which has aroused strong opposition from environmental groups (THE, 2024; Horton, 2021). While promoting offshore wind, Ørsted has also been at loggerheads with unions over pay and faced opposition from parts of the local community, particularly over the potential ecological impact of its projects (RMT, 2022; Ørsted, 2024). These issues highlight the complexity of balancing social responsibility and environmental protection while achieving the Sustainable Development Goals.

6. GBE Development and Policy Recommendations

6.1 GBE Financing Model

The above financing model of GBE will have implications for the stability and sustainability of the development model. During the energy transition, the selection of financing options constitutes an important subject. There are various sources from which GBEs can learn, such as government financing, public-private partnership (PPP), green bonds, etc. First of all, as the initial financial support, government funds can ensure the stable operation of GBE, especially in the early stage of energy

transition, government funds can help promote major infrastructure projects such as nuclear power and wind power. However, depending on government funds might also lead to a financially over-reliant situation. Henceforth, in GBE's development, the institution should ensure that it expands the financing strategies planned for the development process (Heavens et al., 2024).

Second, the PPP model can share risk, introduce funds and technology from the private sector and consider the allocation of resources. Particularly in large-scale infrastructural projects, the PPP model can enable all the partners involved to have equal benefits, hence facilitating the efficiency of the project's implementation. From Vattenfall's experience, GBE should seek to adopt the PPP model to enhance the funding source and implementation capability of projects, according to Zhang (2024).

6.2 GBE Pricing Strategy

GBE's pricing policy should be based on economic advantages and social costs. GBE needs to adopt a reasonable pricing strategy to stabilize the energy supply when the price is volatile due to high market risk. The dynamic price cap is widely used because it enables a company to change the cost of its products based on the conditions experienced in the market. While the fixed form of price cap can only set the maximum price for concerned energy at a fixed ceiling, the dynamic type can naturally be adjusted upward in the course of high demand or low supply to avoid frequent energy price fluctuations and to safeguard the consumers' rights and interests (Edge Branding, 2024). Furthermore, GBE can also present a flexible pricing strategy in which the organization offers different prices based on pricing strategies and market demand. For instance, considering that high electricity tariffs are likely to deepen energy poverty, price subsidies can be given out to low-income households. This way, the social responsibility of the GBE will be enhanced while eradicating energy poverty.

6.3 GBE Investment Priorities

The GBE should focus on key areas such as offshore wind power, grid energy storage and hydrogen energy according to current and future trends in energy development to

achieve sustainable development. Offshore wind power is one of the core directions of the UK's energy transition, and GBE should prioritize investment in offshore wind power projects to take advantage of the UK's rich offshore wind resources and accelerate the deployment of green energy. Based on Vattenfall's success, offshore wind can provide the UK with a long-term stable supply of electricity, while boosting jobs and economic growth (Renewable, 2024).

The development and application of grid energy storage technology will help GBE overcome the intermittency problem of renewable energy and ensure the stability of energy supply. GBE should increase investment in grid energy storage projects, promote breakthroughs in energy storage technology, and provide more efficient scheduling capabilities for future energy systems (ACE Accept, 2020).

As an important part of clean energy, hydrogen energy has great potential for development. The GBE should actively invest in hydrogen production, storage and application technologies to promote the widespread use of hydrogen in industry and transportation. By supporting the development of the hydrogen industry, the GBE will not only be able to meet low carbon emissions targets, but also provide new growth opportunities for the UK economy (SG, 2023).

6.4 Risks to Avoid

In the process of promoting the green energy transition, GBE should be alert to the following potential risks to ensure sustainable development. First, GBE should refrain from an expansion strategy that Ørsted adopted, which is a rapid one. However, Ørsted has been very active in the offshore wind sector, and thanks to the rapid growth of the business, it has experienced specific financial issues and management concerns. Thus, the GBE must adopt a gradual development to ensure both the project's controllability and economic stability so that the dispersion of resources does not become an issue.

Second, the situation also means that the GBE should be aware of possible risks similar to the EDF nuclear waste issue. Despite working in nuclear power for many years, EDF did not find a good way to dispose of nuclear waste. If GBE gets involved

in atomic energy in the future, it has to determine in advance how nuclear waste will be treated and disposed of, as this issue is causing many long-term problems for EDF and the public.

Thus, the future development of GBE is a combination of the capacitive government and market environment and other factors such as the financing model of the company, pricing strategy, investment focus and risk management strategy. About the component policies of the GBE, its appropriate design and targeted deployment could well become a trademark of the system that provides for the green transition of the UK energy sector without compromising stability and sustainability in the supply landscape. For the government formulation of policies, all these aspects should be fully considered; too much dependence should not be made on any single financing scheme, energy prices should be made just and fair, and unreasonable risks should be avoided.

7. Limitation & Conclusion

This study has several limitations in its analysis of GBE (Great British Energy) and its comparison with companies such as EDF, Vattenfall, Ørsted and others in terms of financial, social and environmental responsibility. First, time constraints prevent in-depth analysis of each company's financial data from covering changes over a longer time horizon, potentially affecting the accuracy of trend analysis. Second, due to space constraints, this report does not fully explore all the potential influencing factors, and only selects a few key financial and social environmental indicators for comparison, resulting in a relatively small sample size that does not fully consider the comparative analysis of more energy companies or other relevant cases. In addition, the availability of data is also an issue, and some companies may not disclose certain details of the data, which affects the comprehensiveness of the data and the breadth of the research. Finally, uncertainty about government support and policy changes may also affect the outlook for the GBE, factors that are not sufficiently discussed in this report.

Understood as a new phase of the Energy transition in the center of energy social innovation project, the foundation of Great British Energy (GBE) represents the innovative effort of a specific model of public energy governance, which has the double objective of supporting the green energy development whilst nurturing economic stability and social justice. The lessons GBE can learn include green energy investment, financing model, governance structure and social responsibility from three state-owned energy companies: EDF, Vattenfall and Ørsted. The following have shown the capability of balancing environmental goals and sound operational balances through significant investments in renewable sources of power, flexible financial options and a balanced portfolio of energy sources.

However, as with any learning, GBE must also remember that one has to learn by applying the institutional constraints and market conditions that might be faced by international experience during the localization process. The UK energy market is highly privatised and consumers are sensitive to price changes, so the GBE needs to keep energy prices affordable while promoting green investment. To this end, the government can ease the financial pressure by introducing diversified financing channels such as green bonds, government capital injection and PPP models, and adjust the adverse impact of market fluctuations on consumers through dynamic price ceiling mechanisms.

Looking ahead, the GBE will not only continue to play a leading role in promoting technological innovation and green energy development, but also play a national responsibility in equitable energy access, energy poverty alleviation and ecological protection. In the context of the decarbonization of the energy system, the GBE's mission is to become much more than a public enterprise, but also a key driver of the UK's Sustainable Development Goals and energy justice transition. Only by forming a positive interaction between institutional design, governance efficiency and social identity can GBE move forward steadily in the future turbulent global energy landscape.

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